

NORTHWESTERN UNIVERSITY
IEMS 404-1: Predictive Analytics I (Parametric Regression and Classification Models), Fall 2025

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apley@northwestern.edu; office hours: Fri 1:00-1:55 M235.

Lecture: M/W 3:30-4:50, TECH M228.

Canvas: Lecture notes, handouts, homework assignments, data sets, announcements, etc, will be posted regularly.

Northwestern University Syllabus Standards: This course follows the [Northwestern University Syllabus Standards](#). Students are responsible for familiarizing themselves with this information.

Prerequisites: IEMS 401 (or a good prior course in statistics) and comfort with linear algebra (matrix/vector algebra, matrix inversion, eigen-decomposition).

Texts:

- (T) *Predictive Analytics: Parametric Models for Regression and Classification Using R*, by Ajit Tamhane, Wiley, 2021.
- (HTF) *The Elements of Statistical Learning: Data Mining, Inference, and Prediction. 2nd Ed.*, by Trevor Hastie, Robert Tibshirani, Jerome Friedman, Springer, 2009. Available for free from author's website <https://web.stanford.edu/~hastie/ElemStatLearn/>. (thorough, expansive coverage of statistical learning, but aimed at a statistically mature audience)
- (JWHT) *An Introduction to Statistical Learning with Applications in R, second edition*, by Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani, Springer Texts in Statistics. **ISBN-13:** 978-1071614174. Available for free at <https://www.statlearning.com/> or https://hastie.su.domains/ISLR2/ISLRv2_website.pdf (you may have to run NU VPN to have access)

Grading: Midterm	25% (Wed 10/29/25, in class)
Final	25% (Wed 12/3/25, in class)
HW	25% (see below)
Project	25% (see below)

Software: Primarily R, which is open source freeware that you can download (along with tutorials, etc) from www.r-project.org.

Course Objectives: This course is the first part of a two-part sequence, IEMS 404-2 being the second course. The availability of data for building predictive models to predict one or more variables as functions of another set of observed variables is exploding. The objective of the two-part sequence is to cover concepts and tools that enable students to skillfully build, interpret, and use predictive models. Whereas 404-2 covers nonparametric and more complex nonlinear modeling methods, 404-1 focuses on classical parametric models (primarily linear and logistic regression, generalized linear models, survival analysis, and clustering), especially the handling of pitfalls that often corrupt fitted predictive models. Although this course is applied in that the emphasis is on how to effectively implement predictive modeling, certain theoretical concepts that help build an understanding of the methods will also be covered.

Course Outline: (Since JWHT targets a broad audience and often lacks technical details, and HTF is often dense and difficult to read, the lecture notes are an important complement)

1. Multiple linear regression: Basics (Notes_1.ppt)
 - The linear regression problem
 - Model fitting – least squares
 - Geometry of least squares
 - Fisher-Cochran Theorem, ANOVA decomposition, and assessing the fit
 - Distribution of regression coefficients and Gauss-Markov Theorem
 - Statistical inference on the regression coefficients – t tests and partial F tests
 - Partial correlation coefficients
 - Model predictions and uncertainty quantification
2. Regression diagnostics and other issues (Notes_2.ppt)
 - Leverage, influence, and outlier detection
 - Recursive least squares
 - Residual diagnostics
 - Heteroskedasticity, variance stabilizing transformations, and weighted least squares
 - Standardized regression
 - Handling categorical predictors
 - Interactions: An important type of nonlinearity
3. Model and variable selection, shrinkage, and multicollinearity (Notes_3.ppt)
 - Model evaluation criteria – C_p , AIC, BIC, $PRESS_p$, Cross-Validation
 - Variable selection: Stepwise and best subsets regression
 - Multicollinearity and multicollinearity assessment
 - Multicollinearity fixes: Ridge and lasso regularization/shrinkage
 - Principal components analysis (PCA) for unsupervised learning
 - Multicollinearity fix: principal components regression (PCR)
4. Logistic regression and generalized linear models (Notes_4.ppt)
 - Categorical responses and logistic regression basics
 - Model fitting: MLE and iteratively reweighted least squares
 - Statistical inference and model selection in logistic regression
 - Diagnostics in logistic regression
 - Multinomial logistic regression (nominal and ordinal)
 - Generalized linear models (GLMs):
 - Exponential families and link functions
 - Model fitting: MLE and iteratively reweighted least squares
 - Poisson regression
 - Gamma regression
5. Survival analysis (Notes_5.ppt)
 - Hazard rate and survival distribution

- Kaplan-Meier estimator
- Cox's proportional hazards model

Exams: The midterm and final will be closed book and closed notes, except you are allowed one page of "cheat sheet" notes (8-1/2" by 11", front and back) for the midterm and two pages for the final exam. If any statistical tables are needed, I will post a pdf of them on Canvas, which you can print out and bring to the exam (no notes written on the back/margins/etc. of these, though). Like the homeworks, the exams will stress implementing the methods in software, in addition to theory and concepts behind the methods. Exam questions on implementing the methods will focus on interpreting the output of various R packages (no coding though). The best way to study for the exams is to make sure you are familiar with all the points from the lecture notes and the homeworks.

Homework: Homework will be assigned roughly biweekly and will be conducted and turned in individually. It will typically be a combination of data-oriented questions and some more theoretical questions. The former will involve analyzing a set of data using the methods covered in lecture and drawing appropriate conclusions and interpretations. Homeworks must be turned in on time to receive credit.

Homeworks will not be graded in any detail. If you turn in the homework solutions on time, you will receive full credit (even if there are mistakes), as long as you make a concerted effort to solve all problems.

Project Teams: The projects will be conducted in teams of 3 students (2 or 4 students may be okay, if there is a compelling reason, but it needs approval from the instructor).

Project: The project will be conducted, presented, and turned in as a team assignment (see project teams above). The project can involve any subset of the methods covered in class, or perhaps related methods if more relevant. The project should revolve around a particular predictive analytics problem and questions that you would like to answer, and the problem should dictate the methods that you use (as opposed to vice-versa). Ideally, this will be your practicum project, but you can choose another relevant analytics problem and data set if appropriate. You may be able to find an appropriate problem and data from one of the publicly accessible data repositories (e.g., <http://archive.ics.uci.edu/ml/datasets.html> [my favorite], <http://www.kaggle.com/competitions>, <http://www.kdnuggets.com/datasets/index.html>, <http://lib.stat.cmu.edu/>, <http://kdd.ics.uci.edu/databases/>, <http://www.statsci.org/datasets.html>, etc). Time permitting, teams will present their results in class, the last week of class.

Final project reports are due Friday, December 5, by 11:59 pm. (post an electronic version on Canvas).

Some guidelines for the projects:

- Make sure to have an interesting problem or question to be answered, and an appropriately sized data set that will help to analyze the problem.

- The project should involve some “response” variable that you want to predict, as a function of a number of other “predictor” variables that will be available at the time of prediction. This is the most generic supervised learning setting and is the primary subject of this course.
- If you obtain your own data from a real source other than some repository, you will almost certainly need to invest significant time in data “pre-processing”, including defining appropriate predictor variables from unstructured data repositories, cleaning bad/missing data, transforming heavy tailed variables, etc. This is important and should be discussed in your report, even though it is not the subject of this course. In particular, you should have a lengthy discussion on why you considered the predictors that you considered.
- Most/all projects will involve finding the best possible supervised learning model, in terms of giving the most accurate response prediction. This will involve trying and comparing different models (e.g., with different combinations of predictors, different regularization, etc). If your conclusion is that there is no response predictability (i.e., if the global average is the best predictor), that is okay. But try to avoid selecting a project for which this is the case.
- For the final project report, include your R code as Appendices in the electronic (pdf) version of your report. But do not include the code in the hard copy printout. For the electronic version, please label the different parts of your code and refer to the labels when discussing the results in the main body of the text.
- A very important part of the final project report is a clear and detailed description of your data: What were your raw data, what exactly was the data array to which you fit the supervised learning model, what does each row correspond to, what was the response variable for each row, what were the predictor variables for each row, etc. It should be described in enough detail that, given your raw data, someone could reconstruct the data array you used for regression/classification.